Machine Learning Algorithms Used in Fraud Detection

Professionals, scholars, and other stakeholders have used a wide range of ML algorithms and approaches in fraud detection. Key among the approaches is logistic regression, which scholars and practitioners, and other stakeholders, apply as a baseline supervised classification algorithm because of its simplicity, computational efficiency, and interpretability (Qi, 2025; Alrasheedi, 2025). According to Chen (2025), the logistic regression algorithm performs well for linearly separable fraud detection problems but struggles to capture complex, nonlinear relationships inherent in modern financial fraud. Evidently, as studies, such as Islam et al. (2024), Hashemi, Mirtaheri, and Greco (2022) demonstrated, Logistic Regression generally achieves lower recall and ROC-AUC than ensemble and deep learning models, particularly on highly imbalanced datasets. However, as Islam et al. (2024) pointed out, despite the numerous limitations associated with Logistic Regression, it is still a critical benchmark for evaluating the more recent and advanced fraud detection models.

Decision Trees

Another ML algorithm used in fraud detection is Decision Trees. Professionals and institutions frequently use Decision Trees because they produce transparent decision rules that the users can easily interpreted (Zhang et al., 2023). In agreement, Du et al. (2024) observed that Decision Trees are effective in modelling nonlinear decision boundaries and require very little data pre-processing. However, according to Chen (2025), individual Decision Trees are susceptible to overfitting, which results in lower predictive performance than ensemble approaches. According to Du et al. (2024), Decision Trees algorithms and models improve considerably when institutions incorporate them within ensemble learning frameworks, such as Random Forest and Gradient Boosting.

Random Forests

The next algorithm that institutions use in fraud detection is Random Forest. The Random Forest algorithm, according to various studies, has demonstrated higher effectiveness in fraud detection, which is attributable to their robustness, high predictive accuracy, and ability to reduce overfitting (Al-dahasi et al., 2025; Hashemi, Mirtaheri and Greco, 2022). In agreement, Islam et al. (2024), Random Forest effectively handle high dimensional datasets, but at the same time, maintain strong performance even in cases with datasets with extreme class imbalances. Similarly, articulating the strengths of Random Forest, Chen (2025) observed that it outperforms many conventional ML algorithms in terms of precision, recall and F1-score. However, Chen (2025) pointed out that the complexity associated with Random Forest reduces its interoperability.

Support Vector Machine (SVM)

Support Vector Machines (SVM) is another ML algorithm that financial institutions and other stakeholders in the sector have used in fraud detection and prevention. SVM, as Zhang et al. (2023) observed, have demonstrated strong performance in fraud detection by effectively separating fraudulent and legitimate transactions within high-dimensional feature spaces. In their publication, Zhang et al. (2023) pointed out that SVM had one of the highest classification accuracy compared to various conventional algorithms based on tests on the BankSim dataset. However, other studies have demonstrated that in cases where there are extreme or severe class imbalances and large transaction datasets, the performance and accuracy of SVM algorithms reduces (Chen, 2025; Eswar Prasad et al., 2023). Therefore, from the literature, in cases where there are moderately sized datasets or there are no extreme class imbalances, SVM models remain one of the most effective algorithms.

K-Nearest Neighbours (KNN)

KNN is another widely used ML algorithm because of its conceptual simplicity and ability to classify transactions based on similarity to neighbouring observations (Zhang et al., 2023). KNN has demonstrated reasonably well performance in cases where the objective is to predict fraud in cases where fraud patterns are clearly distinguishable within the feature spaces. However, Chen (2025) observed that KNN is associated with high computational costs during prediction, and is sensitive to irrelevant features and noisy data. As such, considering the nature of KNN, and the findings related to its application, it is important note that its applicability in real-time fraud detection is extremely limited.

Other Models and Algorithms

There are other numerous models and algorithms that institutions in the financial services sector have used in fraud detection. The Naïve Bayes provides rapid classification with minimal computational requirements, making it attractive for preliminary fraud detection screening (Zhang et al., 2023). However, according to Chen (2025), the assumptions of the model about feature independence is not applicable in financial transaction data, which reduces its predictive performance relative to other more advanced and sophisticated algorithms. The weaknesses associated with the model have resulted in the various industry players considering Naïve Bayes as a useful benchmark, instead of a state-of-the-art fraud detection model.

Another algorithm is the Artificial Neural Networks (ANN), which players in the industry consider as an effective model in cases of complex nonlinear fraud patterns that traditional statistical models cannot capture (Eswar Prasad et al., 2023). Further, according to Islam et al. (2024), ANN models are have demonstrated high recall and precision after appropriate feature engineering and class balancing in datasets. Eswar Prasad et al. (2023) observed that ANN outperformed Decision Trees and SVM after applying SMOTE and dimensionality reduction.

However, as Bello et al. (2023) observed, ANN models require extensive computational resources and large training datasets.

Another recent model or algorithm is the Deep Neural Networks (DNN). The DNN model, according to Alrasheedi (2025) model has demonstrated an ability to improve fraud detection by learning hierarchical feature representations from complex transaction data (Alrasheedi, 2025). Further, according to Islam et al. (2024), DNNs have consistently demonstrated superior performance, outperforming conventional algorithms in cases where there are large volumes of behavioural and transactional data. However, Chen (2025) pointed out that the adoption for DNNS has remained lower compared to other models because of high computational costs, long training times, and limited explainability.

Other models, autoencoders are widely applied for unsupervised anomaly detection where labelled fraud examples are limited (AbouGrad and Sankuru, 2025). The autoencoders algorithms identify fraudulent transactions by reconstructing errors instead of explicit class labels. Considering their characteristics and performance, autoencoders are useful in detecting previously unseen fraud patterns but they exhibit lower recall rates than the supervised learning models (AbouGrad and Sankuru, 2025; Ayeomoni, 2022). Another model, the XGBoost, has demonstrated a high performing fraud detection pattern across multiple studies and reports (Al-dahasi et al., 2025; Islam et al., 2024). The XGBoost algorithm handles nonlinear relationships, missing values, feature interactions, and class imbalance effectively; with some of the highest accuracy, precision, recall, F1-score, and ROC-AUC among different models (Islam et al., 2024). However, Islam et al. (2024) pointed out, the XGBoost algorithm is associated with computational complexity and reduced interpretability.

In another report, Hashemi, Mirtaheri, and Greco, 2022), pointed out another model, LightGBM, provides faster training and prediction compared to various boosting algorithms, while at the same time, maintaining high predictive performance, with some of the most effective performance being noted in scenarios with large-scale transaction datasets because of its efficient tree growth strategy. Notably, studies have demonstrated that LightGBM achieves performance at the same levels as XGBoost, but does not require the same levels of computational times (Ding et al., 2024). Hashemi, Mirtaheri, and Greco (2022) also discussed another algorithm, CatBoost, which improves fraud detection by efficiently processing categorical transaction variables without extensive pre-processing. Users have reported that CatBoost performs competitively compared to XGBoost and LightGBM algorithms, but there is inadequate empirical evidence on its performance in banking fraud detection contexts (Theodorakopoulos et al., 2025).

Gradient Boosting Machine has higher predictive accuracy through sequential correction of previous classification errors, resulting in better performance compared to traditional and legacy models in classifying fraud, but require careful hyperparameter tuning to prevent overfitting (Dichev, Zarkova and Angelov, 2025). Another algorithm, Isolation Forest, detects anomalies by isolating observations that are substantially different from normal transaction behaviour, with best results evident in situations with unsupervised fraud detection where labelled fraud cases are unavailable, but with lower precision than ensemble supervised algorithms (Du et al., 2024). Ensemble Learning Models combine two or more classifiers to improve predictive performance, robustness, and generalisability, with Random Forest, XGBoost, and voting ensembles, as well as stacking approaches outperforming individual classifiers across multiple evaluation metrics by reducing false positives, but maintaining high fraud detection rates (Chen, 2025; Khalid et al., 2024). However, the increased computational complexity and lower interpretability are key impediments to implementation of the model. Other models fall under hybrid machine learning models, which integrate multiple ML techniques or combine different supervised and unsupervised learning to improve fraud detection (Li and Chen, 2025). Others, including dynamic quantification anti-fraud models and hybrid ensemble architectures have demonstrated improved adaptability to evolving fraud patterns and adversarial evolution (Israfilov, Yatskiv, and Spiridovska, 2025; Li and Chen, 2025; Yang, Shukur, and Sahran, 2026). However, it is noteworthy, as Btoush et al. (2025) observed, although hybrid models achieve superior performance, their implementation complexity and computational requirements remain considerably higher than single-model approaches.